

Review

Aphasia in vascular lesions of the basal ganglia: A comprehensive review

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ABSTRACT

Between 1970 and 1990, the study of aphasia secondary to subcortical lesions (including the basal ganglia – BG) was largely driven by the advent of modern neuroimaging techniques such as MRI and PET. However, attempts to characterize a pattern of language abnormalities in patients with basal ganglia lesions proved unfruitful. We conducted a comprehensive review of **language disturbances after vascular lesions in the BG**. Literature search in Medline and LILACS (1966–2016) and PsychINFO (last 25 years) was conducted, and returned 145 articles, with 57 eligible for the review yielding data on 303 patients. We report the clinical and neuroimaging features of these cases. Results showed that **aphasias caused by BG lesions are heterogeneous with weak clinicoanatomical correlations**. Data derived from follow-up and flow/metabolism studies suggest that subcortical aphasia caused by BG lesions involves hypoperfusion in the cortical territories of the middle cerebral/internal carotid arteries (MCA/ICA) and their branches.

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1. Introduction

Subcortical aphasia is currently defined as aphasia secondary to lesions involving the basal ganglia and thalamus, without

involvement of cortical structures and minimal compromise of connecting white matter pathways (Alexander & Hillis, 2008).

The advent of modern neuroimaging techniques has allowed a more precise visualization of subcortical structures and consequent identification of neurological syndromes in which subcortical involvement is predominant or exclusive. A great interest in the study of subcortical lesions has emerged in Cognitive Neurology, since the notion that subcortical lesions can impair cognitive functions, although long-held, called for a more accurate confirmatory method of anatomic study *in vivo*. Thus, although interest in language disturbances secondary to subcortical lesions began short after the description by Marie (1906), the widespread clinical use of cranial computed tomography (CT) in the 1970s led to a large number of studies focusing on the role of these structures in language processing.

Pioneering studies on the role of subcortical lesions in language usually centered on extensive brain lesions often involving large areas of periventricular white matter (PVWM) and associated cortical lesion. Following the advent of magnetic resonance imaging (MRI), emphasis was placed on studying smaller and more limited lesions to improve anatomic-clinical correlations, but most studies still addressed lesions affecting both the basal ganglia structures and the PVWM, in varying combinations.

Although attempts to characterize subcortical aphasic “syndromes” in a similar way to classic cortical syndromes have proved unfruitful, it is widely accepted that subcortical lesions share some features, such as a more benign clinical picture and better long-term prognosis. Striatocapsular lesions may lead to articulatory and word-finding deficits, and comprehension may be impaired at more complex syntactic levels, but further anatomic-clinical correlations were found to be much less consistent (Alexander & Hillis, 2008; Ardila, 2014).

Currently, language disturbances secondary to lesions in the basal ganglia are attributed mainly to hypoperfusion and associated cortical ischemia. This notion is justified by the fact that most vascular lesions in this region are caused by occlusion of the middle cerebral artery and its branches or by hemorrhages that can exert a mass effect on adjacent language-related cortex (Hillis et al., 2002, 2004). However, arguments supporting a direct role of the basal ganglia in language processing can be found in functional studies (Crosson et al., 2003; Gill Robles, Gatignol, Capelle, Mitchell, & Duffau, 2005).

Our objective was to present a comprehensive review of studies on language disturbances due to single lesions of vascular etiology (strokes) involving exclusively the basal ganglia (and/or adjacent internal capsule), to identify the pattern of these language disturbances, and also to discuss the main physiopathological hypotheses for their occurrence.

2. Methods

2.1. Search strategy and study selection

A search was conducted of the Medline and LILACS databases for the period spanning from 1966 to 2016, and of the PsychINFO database for the last 25 years, including the following MeSH terms: *subcortical*, *basal ganglia*, *putamen*, *caudate nucleus*, *lenticular nucleus*, *striatum* associated through the Boolean terms ‘OR’ and ‘AND’ to *aphasia*, *language*. This preliminary search yielded the following results: “*subcortical aphasia*”: 476 papers; “*basal ganglia aphasia*”: 426 papers; “*putamen aphasia*”: 90 papers; “*caudate aphasia*”: 105 papers; “*lenticular nucleus aphasia*”: 123 papers; “*striatum aphasia*”: 140 papers; *globus pallidus aphasia*: 32 papers; “*basal ganglia language*”: 1348 papers; “*putamen language*”: 241 papers; “*caudate language*”: 375 papers; “*lenticular nucleus*

language”: 401 papers; “*striatum language*”: 535 papers; *globus pallidus language*: 135; performing a total of 4427 papers. Titles and abstracts were screened and those containing the terms *dementia*, *Alzheimer*, *Parkinson*, *Huntington*, *schizophrenia*, *alcoholism*, *cerebellum*, *epilepsy*, *obsessive-compulsive*, *bipolar*, *corticobasal*, *PSP*, *Creutzfeldt-Jacob* were excluded, as were all duplicated papers. The initial search retrieved 1323 articles. Abstracts of all these papers were assessed for content, with a second screening of articles that met the following criteria: studies reporting patients aged older than 18 years, with a single focal vascular lesion (ischemic or hemorrhagic) of subcortical structures (including basal ganglia and internal capsule but not extending to the PVWM) in both hemispheres documented by neuroimaging study and language disturbances secondary to these lesions, published in English, French, Spanish, Italian or Portuguese, and to which authors had access to the full articles. At this point, papers focusing on purely theoretical or experimental aspects of basal ganglia role in language, as well as those focusing on non-vascular lesions or patients up to 18 years of age were excluded. This selection process yielded 145 articles.

The resultant 145 articles were then assessed in full, and a further selection was performed. The primary objective of some studies was not to investigate language disturbances but to provide a general clinical description of “subcortical syndromes”, where no specific language tests were applied. In this second stage, most of the excluded articles comprised: (a) description of language disturbances or neurocognitive experiments involving frontostriatal circuits with associated frontal lesions, (b) perisylvian injuries associated with subcortical lesion, (c) multiple cerebral lesions, (d) studies that did not investigate basal ganglia structures separately, (e) studies that did not provide individual descriptions of patients’ symptoms, or (f) studies in which patients could not be matched with the corresponding lesions on neuroimaging. We also manually searched the reference lists of these articles and selected those that were relevant. This last analysis led to the final inclusion of 57 articles involving a total of 303 patients. In some cases, a few patients in a given study were excluded for failing to meet the above outlined criteria. (Table 1 and Fig. 1).

2.2. Patient selection

Patients aged over 18 years of both sexes with a single (left or right hemisphere) vascular lesion in the basal ganglia and without PVWM or cortical involvement were included. Cases reported to be non-aphasic in the acute stage were included. Out of the total 303 patients included, 6 were subsequently excluded (patients 55, 108, 115, 147, 157, and 269) for not fully satisfying the predefined inclusion criteria, giving a final total of 297 patients (Appendix A). Studies were included independently of how much time had elapsed from the ictus to the actual report. Therefore, we divide our results in: patients with aphasia in the acute stage (up to 30 days following the ictus), subacute stage (from 31 days up to 6 months after the ictus), and chronic (more than 6 months after the ictus). Patients were not enrolled in this review if there was any mention that they had been submitted to speech therapy in the original study.

2.3. Language assessment

All articles for which data on the parameters fluency, comprehension impairment, repetition deficits, presence of anomia, presence of paraphasias (phonemic and semantic) was available were included. Articles containing a description of results obtained on language tests were included, as were those in which disturbances were described by the authors of the study after clinical examination. Reading and writing abilities were not taken into account for

Table 1
Studies included in the review.

Study	N (patients)	Language assessment	Complementary neuropsychological assessment
Cambier et al. (1979)	1	Oral expression, repetition, oral comprehension, reading, writing; Verbal Fluency	Clinical evaluation of calculus, motor praxis
Alexander and LoVerme (1980)	5	BDAE or PICA	WAIS; WMS; nonverbal problem solving
Damasio et al. (1982)	4	Multilingual Aphasia Examination	Performed, but tests not described
Cappa et al. (1983)	7	Oral expression, auditory comprehension, repetition, written expression, reading comprehension, reading aloud, dictation, copying; Token Test	Raven's Progressive Matrices; Weigl's Color-Form Sorting Test
Metter et al. (1983)	2	BDAE; PICA	Verbal and nonverbal memory tests
Wallesch et al. (1983)	16	AAT; BDAE; Token Test; production of synonyms and antonyms; translation of common idiomatic expressions; verbal fluency (Semantic and FAS)	WAIS
Puel et al. (1984)	9	Montréal-Toulouse Aphasia Battery	NP
Stein et al. (1984)	1	Overall clinical evaluation	Overall clinical evaluation
Vergara et al. (1984)	4	Fluency, comprehension, repetition, naming, and presence of paraphasias	Gnosias and praxis
Fromm et al. (1985)	8	WAB	NP
Tanridag and Kirshner (1985)	1	BDAE	NP
Wallesch (1985)	7	BDAE or AAT; Token Test	NP
Lieberman et al. (1986)	2	PICA	NP
Olsen et al. (1986)	17	Spontaneous speech, sentence repetition, object naming and interpretation of proverbs; short Token Test	Clinical evaluation of intellectual functions
Alexander et al. (1987)	1	BDAE	NP
Basso, Della Salla, and Farabolo (1987)	13	BDAE	NP
Mehler (1987)	1	Overall clinical evaluation	Clinical evaluation of memory and visuospatial functions
Perani et al. (1987)	4	Auditory verbal comprehension, oral expression, repetition, reading and writing; Token Test	Neglect battery
Viader et al. (1987)	1	Overall clinical evaluation	Overall clinical evaluation
Vallar et al. (1988)	3	Auditory verbal comprehension, oral expression, repetition, reading and writing; Token Test	Neglect battery
Guarnaschelli, Pistarini, Fugazza, and Baldi (1989)	13	Comprehension (oral and written), expression, repetition, naming, writing; Token Test	NP
Saggese, Taboada, Duhart, and Adaro (1989)	9	Fluency, comprehension, repetition, naming, reading, writing and presence of paraphasias	NP
Caplan et al. (1990)	2	Overall clinical evaluation	Overall clinical evaluation
Robin and Schienberg (1990)	10	BDAE; Token Test; Reading Comprehension Battery for Aphasia	NP
Weiller et al. (1990)	29	AAT	Overall clinical evaluation
De Renzi et al. (1991)	2	Naming, repetition, confrontation naming, comprehension of verbal commands	Cube and box test; Matching to sample test, Assessment of apraxia
Pedraza et al. (1991)	8	Overall clinical evaluation	Overall clinical evaluation
Sonobe et al. (1991)	1	Standard Language Tests of Aphasia	WAIS
Bartolomei, Nicoli, Swiader, and Gastaut (1992)	1	Overall clinical evaluation	NP
Démonet et al. (1992)	5	Montréal-Toulouse Aphasia Battery	NP
Kennedy and Murdoch (1993)	5	WAB; Cookie Theft Picture description; BNT; NCCEA	NP
Willmes and Poeck (1993)	3	AAT	NP
Mega and Alexander (1994)	2	WAB; BNT; Token Test (short version)	WCST
Milhaud et al. (1994)	6	Overall clinical evaluation	Overall clinical evaluation
Pulicino et al. (1994)	1	Overall clinical evaluation; picture naming	Performed, but tests not described
Fuh and Wang (1995)	6	BNT; WAIS: comprehension, word and sentence reading, number reading and writing, verbal fluency subtests	MMSE; WAIS: Digit span, Proteus Maze-like Maze, Delayed Picture Recognition Span, paired word learning, remote memory, simple calculation, pattern-matching, simple figure copying subtests
Fabbro et al. (1996)	1	AAT, Bilingual Aphasia Test	Evaluation of apraxia, visual agnosia, long-term and short-term verbal memory, and intelligence
Giroud et al. (1997)	20	Fluency, repetition, word definition, writing and reading examination	Drawing a cube; orientation on a map; long-term memory, short-term memory and recall; WCST; TMT; Luria's motor series and problems; Rey-Osterrieth figure copy; Stroop test
Halkar et al. (1997)	1	Overall clinical evaluation	NP
Takahashi et al. (1997)	2	Overall clinical evaluation	Overall clinical evaluation
Hochstenbach, van Spaendonck, Cools, Horstink, and Mulder (1998)	12	Similarities subtest (WAIS); Sentence comprehension; Verbal fluency	AVLT; Digit span (WAIS); RBMT; TMT-A and B; Block design (WAIS); Clock drawing; Bobertag; Copying task (BIT); Money's Road Map
Friederici, von Cramon, and Kotz (1999)	3	Sentence comprehension	NP
Kumral et al. (1999)	31	Turkish Aphasia Test	MMSE; RAVLT; BVRT; Stroop test; Luria's test
Carlier et al. (2000)	1	Oral and writing comprehension, naming and repetition	NP
Ciabarra, Elkind, Roberts, and Marshall (2000)	1	Overall clinical evaluation	Overall clinical evaluation
Warren et al. (2000)	1	BDAE; WAB; Mount Wilga High Level Language Test	WAIS III
Hua et al. (2001)	18	Visual naming, object naming, sentence repetition,	NP

Table 1 (continued)

Study	N (patients)	Language assessment	Complementary neuropsychological assessment
Riecker, Wildgruber, Grodd, and Ackermann (2002)	1	semantic association of verbal fluency, Token Test, Aural and Reading Comprehension test, oral reading, spontaneous writing, writing from copy and writing from dictation AAT	NP
Kotz et al. (2003)	4	AAT; Token Test	NP
Kuljic-Obradovic (2003)	15	BDAE; BNT; Token test; Verbal Fluency	MMSE
Radanovic and Scaff (2003)	4	BDAE; BNT; Token Test	NP
Russmann et al. (2003)	13	Overall clinical evaluation	Overall clinical evaluation
Charron, Pluchon, Besson, and Gil (2004)	1	BDAE, animal fluency, letter “M” fluency	TMT – A and B; WCST; Stroop test
Radanovic et al. (2004)	7	BDAE; BNT; Token Test; Frontal Scripts	BVRT; WCST; TMT A and B
Troyer, Black, Armilio, and Moscovitch (2004)	1	BNT; Verbal fluency (Semantic and FAS); WAIS: comprehension subtest; Alternate Uses	WAIS: abstraction subtests; Gorham’s Proverbs; Visual-Verbal Test; Rey-Osterrieth Complex Figure; Matrix Reasoning; WCST; Stroop test; TMT A and B; WMS-R, Paced Auditory Serial Addition Test; CVLT
de Boissezon et al. (2005)	5	Montreal-Toulouse Aphasia Battery	NP
Garcia-Caballero et al. (2007)	1	Bilingual Aphasia Test	Addenbrooke’s Cognitive Examination
Krishnan, Tiwari, Pai, and Rao (2012)	7	WAS	NP

BDAE: Boston Diagnostic Aphasia Examination; BNT: Boston Naming Test; PICA: Porch Index of Communicative Abilities; AAT: Aachen Aphasia Test; WAB: Western Aphasia Battery; (R)AVLT: (Rey) Auditory – Verbal Learning Test; RBMT: Rivermead Behavioural Memory Test; BIT: Behavioural Inattention Task; TMT: Trail Making Test; WCST: Wisconsin Card Sorting Test; NCCEA: Neurosensory Center Comprehensive Examination of Aphasia; WAIS: Wechsler Adult Intelligence Scale; WMS-R: Wechsler Memory Scale – Revised; CVLT: California Verbal Learning Test; BVRT: Benton Visual Retention Test; NP: not performed.

the analyses of language disturbances. The disturbances presented by the patients are described as reported in the original articles. Given the heterogeneity in clinical descriptions, the neurolinguistic alterations selected for analysis in this study were based on Crosson’s model of main disturbances expected in subcortical lesions (fluency, comprehension, repetition, naming, presence of paraphasias). Speech disturbances, such as dysarthria, stuttering, and hypophonia were also included in our analysis given that dysarthria (articulatory impairment) are frequent in aphasic syndromes very frequently, especially in anterior (frontal) lesions affecting Broca’s area and its adjacencies; stuttering (a disfluency) has been linked to basal ganglia lesions in the literature; and hypophonia (motor vocal impairment) is a common symptom in Parkinson’s disease, thus sharing an anatomical substrate with some lesion sites focused in this review. Complementary neuropsychological examination is displayed in Table 1.

2.4. Neuroimaging data

Only articles featuring neuroimaging scans (cranial CT, SPECT, PET or magnetic resonance) were included. Studies employing functional methods (cerebral blood flow, SPECT and PET studies) were used for the analysis of correlation between the existence/severity of aphasia and degree of hypoperfusion and for correlation with aspects related to the prognosis of language disturbances.

After the final selection of papers, a database was created to include demographic, clinical, neuropsychological, language and neuroimaging data for all patients reported in each study. All available information was then analyzed by both authors (M.R. and L.L.M.) in order to check for the eligibility of each patient; at this point, 6 patients were excluded and patients were then divided according to lesion side (right or left) and lesion site, as well as time from ictus to language assessment (acute (less than 1 month), 1–6 months and more than 6 months).

3. Statistical analysis

Due to the high heterogeneity of data regarding symptoms description and batteries used in the diagnosis of aphasia, the Chi-square test was applied to assess the association between

lesion sites defined *a priori* and aphasia symptoms and also to determine odds ratio. A significance value of $p < 0.05$ was used for all analyses.

4. Results

Papers included in this review were published between 1979 and 2012. The number of patients per study ranged from 1 to 31. Two hundred and ninety-seven patients were selected for this review, from both sexes, with ages ranging from 21 to 94 years.

4.1. Right hemisphere lesions

Acute phase – Out of the total 297 patients, 72 (24.2%) had right hemisphere lesions (RHL). Of these 72, 57 (79.1%) tested normal for language functions. Disturbances in speech or language were present in 15 (20.9%) patients, comprising 8 (11.1%) cases of pure dysarthria (4 in putaminal lesions, 2 in capsular lesions and 2 in striatocapsular lesions). Four (5.5%) patients with putaminal lesions had repetition impairment. In 2 left-handed patients with striatocapsular lesion, 1 presented with Broca’s aphasia and the other had anomia, comprehension deficit, paraphasia and dysarthria. One right-handed patient with lenticular lesion presented transcortical sensory aphasia.

1–6 months after ictus – Follow-up information for this period was available in 6 cases: 2 patients with lesion in internal capsule were normal; in 3 patients with striatocapsular lesion, 2 had only dysarthria and 1 had anomia; 1 patient with a lenticulocapsular lesion was normal.

More than 6 months after ictus – Follow-up information for this period was available in 9 cases: 6 were normal (putaminal lesions). One patient with striatocapsular lesion had dysarthria. One patient with lenticulocapsular lesions was normal, while another one had impaired comprehension and repetition (Table 2).

4.2. Left hemisphere lesions

Acute stage – Out of the 297 patients selected, 225 (75.7%) had left hemisphere lesions (LHL), and language assessment in the acute stage was described in 180 cases. Of these 180 cases, 71

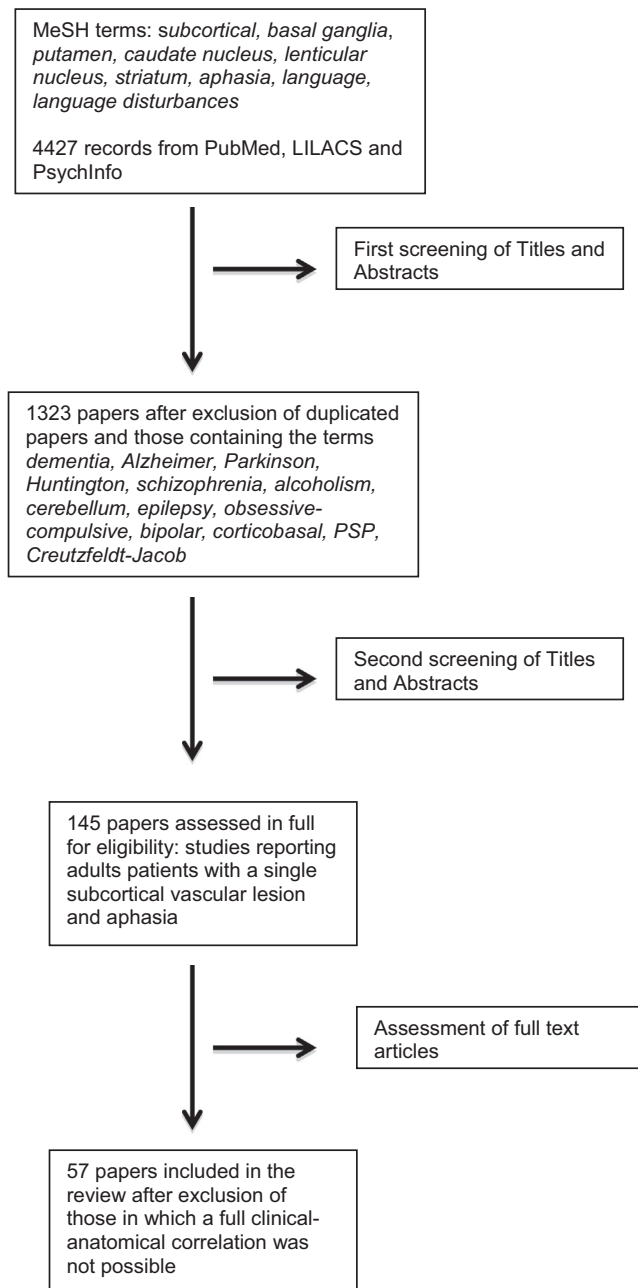


Fig. 1. Selection of articles.

(32.7%) tested normal for language. Language symptoms and their frequencies in this sub-sample are displayed in Table 3.

1–6 months after ictus – Follow-up information for this period was available in 69 cases: 10 patients (14.7%) were normal and the frequency of each language symptom in this sub-sample is displayed in Table 4.

6 months or more after ictus: Follow-up information for this period was available in 39 cases: 10 patients (25.6%) were normal; the frequency of occurrence of language symptoms in this subsample is displayed in Table 5.

4.3. Association between language symptoms and lesion site

Acute stage: patients with lesions in the lentiform nucleus had a higher chance of the occurrence of repetition impairment (OR = 5.78), comprehension deficit (OR = 3.50) and non-fluent

aphasia (OR = 3.23), but were also associated with the absence of language disturbances (OR = 2.65). The occurrence of paraphasia was more strongly associated with putaminal lesions (OR = 2.71), whereas striatal lesions were more strongly associated with the occurrence of dysarthria (OR = 3.67). Non-fluent aphasia was less encountered in patients with caudate lesions (OR = 0.28), and capsular lesions (OR = 0.05). Repetition impairments were less associated with caudate lesions (OR = 0.17). Lesions in the globus pallidus and striatum were not associated with language symptoms (Table 6).

Chronic stage: patients with lesions in the putamen had a higher chance of the occurrence of dysarthria (OR = 17.14) whereas internal capsule lesions were less associated with normality (OR = 0.13) (Table 6).

5. Discussion

The discussion on the role of subcortical structures in language started some 150 years ago when Broadbent (1872) proposed that the basal ganglia “produced” words in the same way as they did with motor actions. Kussmaul (1877) attributed a purely motor role to the basal ganglia as did Marsden (1982) one century later. Carl Wernicke (1874) and Lichtheim (1885) argued that subcortical lesions caused language disturbances only when there was interruption of the connecting pathways among the cortical centers of language, thereby precluding direct participation of the basal ganglia and thalamus in cognitive functions. Most studies of language disturbances due to subcortical lesions center on the theories of Pierre Marie (1906) on speech disorders secondary to lesions in the caudate nucleus, putamen, internal capsule and thalamus, which he termed “anarthria” (a disorder in initiation and control of articulatory movements for speech production), and also on the work of Von Monakow (1914), who implicated lentiform nucleus lesions in the genesis of aphasia. Studies conducted in patients submitted to stereotactic surgery for Parkinson’s disease led to a conceptual revision of the classic models of language-related areas, following the observation that lesion and stimulation of the thalamus or globus pallidus promoted language disturbances during and after the surgical procedures (Ojemann, 1976).

The major advancement in the study of subcortical lesions occurred after the emergence of modern methods of neuroimaging, which improved the establishment of the relationship between specific subcortical lesion sites and neuropsychological deficits, such as aphasic disturbances in lesions of the dominant hemisphere. A large number of studies have been published in recent decades, furthering understanding on the role of subcortical structures in language.

Several studies conducted during the 1960s, 1970s and 1980s implicated the basal ganglia in language, and the first attempts to explain their role in speech and language processing were based on the models of motor function taken from neurophysiology studies and which emphasized the role of these nuclei in the generation of accurate sequential motor programs. Massive connections of the basal ganglia with most cortical regions also pointed to the involvement of these structures in cognitive functions.

The mechanisms that have been proposed to explain the role of basal ganglia in aphasias along decades can be grouped into five main categories, as follows:

1. **Disconnection among cortical areas involved in language:** The first model of the involvement of subcortical structures in the pathogenesis of aphasia attributed an indirect role to these structures: a mechanism of interruption of the connecting pathways between classic areas of language (Wernicke’s area and Broca’s area), a theory advocated by both Wernicke

Table 2

Speech and language symptoms in right hemispheric lesions.

	Normal	Non fluent	Fluent	Comprehension deficit	Repetition deficit	Naming deficit	Paraphasias	Dysarthria	Total
<i>Acute stage (72 cases)</i>									
Putamen	17		3		3			4	24
Putamen plus IC, claustrum or EC	1		1		1				2
Caudate nucleus	13								13
Caudate nucleus plus IC	5								5
Globus pallidus	2								2
Internal capsule	1							2	3
Striatocapsular	17	1	1	1	1	2	1	3	21
Lentiform nucleus plus IC or EC	1		1	1		1			2
<i>1–6 months after ictus (6 cases)</i>									
Internal capsule	2								2
Striatocapsular			1			1		2	3
Lentiform nucleus plus IC or EC	1								1
<i>More than 6 months after ictus (9 cases)</i>									
Putamen	5								5
Putamen plus IC	1								1
Striatocapsular								1	1
Lentiform nucleus plus IC	1		1	1	1				2

IC: internal capsule; EC: external capsule; NR: not reported.

Table 3

Speech and language symptoms in 180 reported cases of left basal ganglia lesion in the acute stage (within 1 month from ictus).

	Normal	Non fluent	Fluent	Comprehension deficit	Repetition deficit	Naming deficit	Paraphasias	Stuttering	Dysarthria	Hypophonia	Total
Putamen	6 (19.3%)	13 (41.9%)	7 (22.5%)	6 (19.3%)	9 (29%)	20 (64.5%)	8 (25.8%)		8 (25.8%)	2 (6.4%)	31
Putamen plus IC or EC	4 (33.3%)	4 (33.3%)	3 (25%)	6 (50%)	4 (33.3%)	6 (50%)	5 (41.6%)		4 (33.3%)	1 (8.33%)	12
Caudate nucleus	14 (58.3%)	3 (12.5%)	6 (25%)	3 (12.5%)	3 (12.5%)	9 (37.5%)	3 (12.5%)		2 (8.3%)		24
Caudate nucleus plus IC or EC	8 (66.6%)	2 (16.6%)	2 (16.6%)	3 (25%)	1 (8.3%)	5 (41.6%)	1 (8.3%)	1 (8.3%)		1 (8.3%)	12
Globus pallidus	4 (100%)										4
Globus pallidus plus IC		2 (66.6%)	1 (33.3%)			3 (100%)			1 (33.3%)		3
Striatum	2 (66.6%)							1 (33.3%)	1 (33.3%)		3
Striatocapsular	11 (32.3%)	12 (35.3%)	6 (17.6%)	11 (32.3%)	12 (35.3%)	16 (47%)	11 (32.3%)	1 (2.9%)	8 (23.5%)	6 (17.6%)	34
Lenticular	3 (20%)	9 (60%)	3 (20%)	7 (46.6%)	10 (66.6%)	9 (60%)			1 (6.6%)		15
Lentiform nucleus plus IC or EC	6 (28.5%)	9 (42.8%)	3 (14.3%)	11 (52.3%)	7 (33.3%)	12 (57.1%)	7 (33.3%)		3 (14.3%)	2 (9.5%)	21
Striatopallidal	1 (100%)										1
Striatopallidal plus IC or EC		3 (100%)		2 (66.6%)		3 (100%)	2 (66.6%)		1 (33.3%)		3
Internal capsule	12 (70.5%)		1 (5.8%)	3 (17.6%)	2 (11.7%)	3 (17.6%)			3 (17.6%)		17
Total	71 (39.4%)	57 (31.6%)	31 (17.2%)	50 (27.7%)	47 (26.1%)	84 (46.6%)	37 (20.5%)	3 (1.7%)	33 (18.3%)	12 (6.6%)	180

IC: internal capsule; EC: external capsule.

(1874) and Lichtheim (1885). This concept was revisited by Alexander, Naeser, and Palumbo (1987), who noted that lesions limited to the striatum or internal capsule (anterior limb) did not produce aphasic symptoms and, even in severe cases, promoted only mild disturbances in naming and stuttering. However, more extensive lesions also affecting the periventricular white matter (anteriorly or posteriorly), temporal isthmus, insula or external capsule, in varying combinations, led to a more consistent pattern of speech and language disturbances. Interruption of pathways connecting the medial geniculate body and the temporal isthmus, and of the callosal auditory pathways in periventricular white matter, explained the comprehension deficits in patients with lesions at these sites. Lesion in the external capsule, extreme capsule and arcuate fasciculus appeared to be critical for the emergence of repetition impairment and phonemic paraphasias. Speech disturbances were more often seen in lesions of the superior periventricular white matter and genu of the internal capsule where descending

corticobulbar fibres are found. Lesions in antero-superior periventricular white matter cause interruption in the pathways between the supplementary motor area and Broca's area leading to transcortical motor aphasia. In addition, anterior callosum pathways can be compromised when there is involvement of the frontal white matter, genu of the corpus collosum and superior periventricular white matter.

Naeser, Palumbo, Helm-Estabrooks, Stiassny-Eder, and Albert (1989) also emphasized the role of white matter damage as a determining factor in the severity of nonfluency, particularly due to interruption of the connections involved in the initiation and preparation of speech movements and their limbic aspects (lesion in medial longitudinal fasciculus), besides impairment in motor execution and sensory feedback to the cortical area of the mouth (lesion in periventricular white matter adjacent to the body of the lateral ventricle, below the sensorimotor cortex corresponding to the mouth).

Table 4

Speech and language symptoms in 68 reported cases of left basal ganglia lesion 1–6 months after ictus.

	Normal	Non fluent	Fluent	Comprehension deficit	Repetition deficit	Naming deficit	Paraphasias	Stuttering	Dysarthria	Hypophonia	Total
Putamen	1 (10%)	2 (20%)	4 (40%)	4 (40%)	1 (10%)	6 (60%)	4 (40%)		4 (60%)	4 (40%)	8
Putamen plus IC or claustrum	1 (8.33%)	7 (58.3%)	3 (25%)	7 (58.3%)	7 (58.3%)	4 (33.3%)	4 (33.3%)		9 (75%)	5 (41.6%)	12
Caudate nucleus	1 (14.2%)	2 (28.5%)	2 (28.5%)	1 (14.2%)	1 (14.2%)	2 (28.5%)	1 (14.2%)				7
Caudate nucleus plus IC	1 (33.3%)	1 (33.3%)	1 (33.3%)	1 (33.3%)		2 (66.6%)	1 (33.3%)		1 (33.3%)		3
Internal capsule	1 (50%)		1 (50%)			1 (50%)					2
Striatum		1 (100%)		1 (100%)	1 (100%)	1 (100%)	1 (100%)				1
Striatum plus claustrum and EC		1 (50%)	1 (50%)	1 (50%)	1 (50%)	2 (100%)					2
Striatocapsular	5 (27.7%)	4 (21%)	4 (21%)	6 (31.5%)	3 (15.7%)	9 (47.3%)	4 (21%)		4 (21%)	3 (15.7%)	19
Lenticular	1 (10%)	3 (30%)	3 (30%)	2 (20%)	3 (30%)	3 (30%)	2 (20%)		1 (10%)		10
Lentiform nucleus plus IC or EC		1 (100%)		1 (100%)	1 (100%)						1
Striatopallidal			1 (100%)	1 (100%)	1 (100%)	1 (100%)	1 (100%)				1
Striatopallidal plus IC		2 (66.6%)	1 (33.3%)	2 (66.6%)	1 (33.3%)	2 (66.6%)	1 (33.3%)	1 (33.3%)			3
Total	10 (14.7%)	22 (31.9%)	21 (31.9%)	27 (39.1%)	20 (28.9%)	33 (47.8%)	19 (27.5%)	1 (1.4%)	19 (27.5%)	12 (17.4%)	69

IC: internal capsule; EC: external capsule.

Table 5

Linguistic features in 39 reported cases of left basal ganglia lesion 6 months or more after ictus.

	Normal	Non fluent	Fluent	Comprehension deficit	Repetition deficit	Naming deficit	Paraphasias	Stuttering	Dysarthria	Hypophonia	Total
Putamen	5 (38.4%)	3 (23%)	3 (23%)	6 (46.1%)	5 (38.4%)	7 (63.6%)	2 (15.3%)		2 (15.3%)	1 (7.7%)	11
Putamen plus IC, EC or claustrum	1 (100%)		1 (100%)	1 (100%)	1 (100%)						2
Caudate nucleus	1 (100%)										1
Caudate nucleus plus IC			2 (100%)	2 (100%)	1 (50%)	1 (50%)	1 (50%)				2
Internal capsule			2 (100%)	1 (50%)	2 (50%)	1 (50%)	1 (50%)				2
Striatum		1 (33.3%)	2 (66.6%)	2 (66.6%)	3 (100%)						3
Striatum plus claustrum or EC	2 (100%)										2
Striatocapsular		3 (60%)	1 (20%)	3 (60%)	1 (20%)	4 (80%)			1 (20%)		5
Lenticular		1 (20%)		1 (20%)		5 (100%)	1 (20%)				5
Lentiform nucleus plus IC	1 (33.3%)	1 (33.3%)		2 (66.6%)	2 (66.6%)	1 (33.3%)	1 (33.3%)				3
Striatopallidal plus IC or EC		1 (33.3%)	1 (33.3%)	3 (100%)	3 (100%)	1 (33.3%)	1 (33.3%)	1 (33.3%)			3
Total	10 (25.6%)	10 (25.6%)	12 (30.7%)	21 (53.8%)	17 (43.5%)	22 (56.4%)	7 (17.9%)	1 (2.5%)	3 (7.7%)	1 (2.5%)	39

IC: internal capsule; EC: external capsule.

2. *Diaschisis*. Perani, Vallar, Cappa, Messa, and Fazio (1987) associated the neuropsychological symptoms of their patients (aphasia or neglect) with cortical hypoperfusion disclosed by SPECT rather than with the anatomical lesion on cranial CT. In a study of 6 aphasic patients and 2 patients with neglect with subcortical lesions submitted to SPECT, Vallar et al. (1988) proposed that the determining factor for the development of neuropsychological disturbances was the presence of hypoperfusion in the ipsilateral cortex, given that patients without cognitive symptoms generally had smaller subcortical lesions and did not exhibit cortical hypoperfusion.
3. *Lesion in subcortical structures directly involved in language*: the study of pure lesions affecting a small portion of subcortical structures, or only a single structure, is a challenge in clinical practice. This difficulty is partly due to the small size of the structures involved (where a few millimeters can represent a huge difference in terms of anatomical location), and partly due to the peculiarities in vascularization of the region, which render lesion of individual structures unlikely. With the possible exception of lacunar infarctions, there is always compromise of different anatomical sites. Nevertheless, some authors have investigated the possible role of the basal ganglia in the genesis of aphasia by circumscribing lesion as far as possible. Damasio, Damasio, Rizzo, Varney, and Gersh (1982) described 9 patients with infarcts involving the head of the caudate

nucleus, putamen and anterior limb of the internal capsule in the dominant hemisphere. The authors found that the majority of patients shared the same features of dysarthria and dysprosody, non-fluent aphasia (only one case resembled Wernickés aphasia) and rapid recovery from symptoms. The authors postulated that lesions in the striatum, affecting its function of programming movement and organizing perception, explained symptoms such as dysarthria and dysprosody, and also subsequent auditory comprehension disturbances. Lesions in the anterior limb of the internal capsule interrupt several pathways critical for language production, such as projections from the frontal cortex to the pons, from the auditory cortex to the head of the caudate nucleus, projections from the thalamus to the motor cortex, and from the dorsomedial thalamus to the pre-frontal cortex, also contributing to the presence of dysarthria, paraphasias and auditory comprehension deficits.

Naeser et al. (1982) proposed the existence of three aphasic syndromes after lesions involving the putamen and internal capsule, as follows:

- (a) putamino-capsular lesion with antero-superior extension (extension in terms of adjacent involvement of periventricular white matter) – patients exhibit predominantly articulatory impairments, with naming deficits yet good semantic

Table 6

Association between language symptoms and lesion site.

Symptom	Putamen		Caudate		Globus pallidus		Striatum		Lenticular		Striatopallidal		Internal capsule	
	p	OR	p	OR	p	OR	p	OR	p	OR	p	OR	p	OR
	(95% CI)		(95% CI)		(95% CI)		(95% CI)		(95% CI)		(95% CI)		(95% CI)	
<i>Acute stage</i>														
Normal	NS		NS		NS		NS		0.047	2.65	NS		NS	
									(1.01–6.98)					
Non-fluency	NS		0.028	0.28	NS		NS		0.005	3.23	NS		0.047	0.05
			(0.09–0.87)						(1.41–7.41)				(0.01–0.91)	
Comprehension	NS		NS		NS		NS		0.003	3.50	NS		NS	
									(1.52–8.05)					
Repetition	NS		0.019	0.17	NS		NS		0.0001	5.78	NS		NS	
			(0.04–0.74)						(2.45–13.64)					
Naming	NS		NS		NS		NS		NS		NS		NS	
Paraphasias	0.014	2.71	NS		NS		NS		NS		NS		NS	
	(1.22–6.01)													
Dysarthria	NS		NS		NS		0.003	3.67	NS		NS		NS	
							(1.55–8.68)							
<i>Chronic stage</i>														
Normal	NS		NS		NP		NS		NS		NS		0.028	0.13
													(0.01–1.04)	
Dysarthria	<0.0001	17.14	NS		NP		NS		NS		NS		NS	
	(5–58.68)													

Chi-square test. Values expressed as significance level (p) and Odds Ratio (OR). CI: Confidence Interval; NS: non significant; NP: not performed. Isolated lesions or those in combination with adjacent white matter structures were included in the analysis (e.g. putamen + putamen with associated lesion of the internal capsule and/or external capsule). Isolated lesions of the internal capsule were analysed separately).

comprehension; periventricular white matter lesion was regarded by the authors as the core element for these findings.

- (b) putamino-capsular lesion with posterior extension – patients presented fluent and paraphasic discourse and severe comprehension deficit; in these cases, the lesion extended toward the auditory radiations in the temporal isthmus, as well as to the globus pallidus, caudate nucleus, external capsule, extreme capsule, claustrum and insula.
- (c) putamino-capsular lesion with antero-superior and posterior extension, resulting in global aphasia. The lesion extended to the globus pallidus, periventricular deep white matter, Broca's area and the temporal isthmus, leading to isolation of Broca's area.

The attempt to establish a correlation between anterior lesions (putamen and anterior limb of the internal capsule) with non-fluent aphasia, and between posterior lesions (putamen and posterior limb of the internal capsule) with fluent aphasia was made by Cappa, Cavallotti, Guidotti, Papagno, and Vignolo (1983). Mega and Alexander (1994) studied the profile of aphasias found in striato-capsular lesions, highlighting the presence of impaired language production (verbal fluency, discourse) with anomia but preserved comprehension and repetition, which suggested a specific deficit in lexical selection. Compromised fronto-striatal systems in the dorso-lateral aspects of the frontal cortex or striatum and their connections to the frontal white matter were thought to underlie these findings.

Much criticism has raised regarding the study of hemorrhagic lesions, based on the argument that the mass effect precludes the establishment of any meaningful anatomo-clinical correlation, yet several studies have pointed the persistence of alterations resembling those described for infarcts when involving the same residual lesion sites (Cappa et al., 1983; D'Esposito & Alexander, 1995; Robin & Schienberg, 1990).

4. *Deficits in regulating the release of language segments formed at the cortical level:* Crosson (1985) proposed that the basal ganglia influence the tonus of the anterior language areas, regulating the flow of excitatory impulses from the anterior ventral

nucleus of thalamus. Fluent aphasias would thus result from an absence of inhibition of the globus pallidus on the thalamus, releasing cortical activation; non-fluent aphasias would result from lesions in the caudate nucleus, whose function is to inhibit the globus pallidus: the lesion may promote over-inhibition of the globus pallidus on the thalamus, leading to reduction in cortical activity. In addition, language segments may be initially formed and subsequently “stored” for later semantic checking by posterior (temporoparietal) mechanisms employed for decoding language. The role of the basal ganglia would be to provide a signal initiating the release of language segments for subsequent motor programming, provided it had been semantically checked by the thalamic circuit outlined above. This theory does not attributes any language processing functions to the basal ganglia that would rather trigger the release of language segments for motor programming. Wallesch and Papagno (1988) suggested that the basal ganglia are involved in the monitoring of multiple lexical alternatives generated in the cortex for selection of the most suitable option for the semantic and motivational requirements.

5. *Current perspectives on subcortical aphasias due to basal ganglia lesions: the hemodynamic factor.* A series of studies in cerebrovascular disorders, particularly by Weiller et al. (1993) provided an alternative physiopathological explanation for subcortical aphasias involving the basal ganglia and adjacent white matter pathways. Based on the premise that striatocapsular lesions are caused by occlusion of the initial segment of the dominant middle cerebral artery (MCA), or occasionally of the internal carotid artery (ICA), the presence of aphasia after lesions of the basal ganglia would be determined by late recanalization of the MCA or poor leptomeningeal anastomoses leading to a more severe degree of neuronal death. Thus, circulatory dynamics is thought to play a crucial role in the manifestation of language symptoms and in the degree of recovery after lesions in the basal ganglia, which in turn are determined by the extent of cortical neuronal loss after prolonged occlusion of the MCA.

Nadeau and Crosson (1997) revisited this theory in greater detail, pointing out that the occlusion of the initial segment of the MCA or ICA impact the entire vascular territory of the former.

This territory includes the perisylvian cortex, essential for language functions. Even in the presence of adequate anastomotic circulation, MCA occlusion often prevents collateral circulation from supplying the infarcted subcortical territory. As consequence, cortical circulation is partially reestablished, preventing the occurrence of a major cortical infarction, although transient neuronal dysfunction may occur, which is sufficiently severe to cause language symptoms, albeit without structural lesion detectable on neuroimaging. In cases of infarctions affecting the posterior limb of the internal capsule with temporal involvement, where the anterior choroidal artery territory is affected, the authors claim that language disturbances are due to a disconnection between thalamus and cortex leading to thalamic aphasia. In hemorrhages of the basal ganglia, the authors also noted circulatory abnormalities that affect the cortex (cortical dysfunction induced by adjacent pressure). The authors concluded that **any role of the dominant basal ganglia in language is obscured by associated cortical dysfunction, being either minimal or non-existent.**

In this review, most patients (79.1%) with right-hemispheric lesions did not exhibit language or speech disturbances while 11.1% of cases had only dysarthria after lesions in the putamen, internal capsule and striatocapsular region. These disturbances may be attributed to the role of these structures in the motor circuit (Tremblay, Worbe, Thobois, Sgambato-Faure, & Féger, 2015). Hua, Chen, and Chu (2001) described four right-handed Chinese patients with right putaminal lesion and repetition deficit. The authors focused on writing dysfunction in subcortical lesions and therefore did not discuss the repetition impairments further, providing no hypothesis for this finding. These patients did not present any other language disturbances (besides writing impairments) or dysarthria. Weiller, Ringelstein, Reiche, Thron, and Buell (1990), and Mega and Alexander (1994) described two patients with right striatocapsular lesion leading to Broca's aphasia and anomia with comprehension deficits, respectively, but both individuals were left-handed. Fromm, Holland, Swindell, and Reinmuth (1985) described a right-handed patient with a right hemorrhagic lentiform lesion, classified as having transcortical sensory aphasia in the acute phase (16 days after ictus), but testing normal on language exams. At the one-year follow-up exam, the authors did not present any hypothesis for the occurrence of aphasia in the acute stage. Residual language disturbances after lesions in right basal ganglia were reported in 2 patients, consisting of anomia (striatocapsular lesion) (Mega & Alexander, 1994), and comprehension and repetition deficits (lesion in lentiform nucleus and internal capsule) (Radanovic, Mansur, Azambuja, Porto, & Scaff, 2004). Mega and Alexander's patient was left-handed, while the patient described by Radanovic et al. (who presented very mild deficits on the Token Test, but not on the BDAE) had hypoperfusion in the fronto-temporo-parietal region as evidenced by SPECT. Hence, it is reasonable to assume that the mild residual comprehension deficits exhibited by the latter patient were due to cortical involvement of areas related to executive and visuospatial functions, impacting performance on the Token Test.

In the case of left basal ganglia lesions, 39.4% of the patients assessed in the acute phase displayed no language disturbances. Approximately 30% of the patients exhibited comprehension deficit while 26% had repetition deficit. Almost half of all cases exhibited anomia (46.6%). Other frequent findings were paraphasias and dysarthria (20.5% and 18.3%, respectively); all types of paraphasia (semantic, phonemic, verbal) were reported in the studies, but such discrimination was done in very few patients, thus hampering further analysis on this symptom. Among patients who remained symptomatic for up to 6 months after ictus, 39.1% had comprehension deficits, 28.9% repetition deficits, and 47.8% anomia. Around 28% of the patients had dysarthria. Of those patients who remained symptomatic for more than 6 months after

ictus, 53.8% had comprehension deficits, 43.5% repetition deficits, and 56.4% anomia.

The lentiform nucleus was found to be associated with the presence of non-fluent aphasia (present in over 50% of cases with this lesion), comprehension impairments, and repetition deficits (48.5% each). However, this lesion site also showed the strongest association with the absence of language disturbances, albeit marginally significant ($p = 0.047$) (Table 5). In 35 cases of acute lentiform nucleus lesions, 26 (74.2%) were classified as aphasic. Among the aphasic patients with a follow up (13), deficits persisted at follow-up in only 5. Two of these patients had occlusion of the branches of the MCA evidenced on arteriography and reduced blood flow in classic language areas (Olsen, Bruhn, & Öberg, 1986). Another patient, who presented impaired verbal fluency, repetition deficit and semantic difficulties in the acute stage, persisted with only residual apraxia of speech (a motor speech disorder leading to phonemic segmental changes and articulatory distortions). The authors stated that this case illustrated the role of the basal ganglia in speech planning, word retrieval and verbal short-term memory (Warren, Smith, Denson, & Waddy, 2000). Unfortunately, no follow-up was available for the majority of the cases of lentiform nucleus hemorrhage. Some studies in which cerebral perfusion was assessed (by SPECT), drew on hypoperfusion findings in language-related cortical regions to explain the presence and persistence of deficits in aphasic patients (Démonet et al., 1992; Perani et al., 1987; Vallar et al., 1988). Metter et al. (1983) e Metter (1992) described frontal and temporoparietal hypometabolism in extensive lesions of the lentiform nucleus and internal capsule. Russmann, Vingerhoets, Ghika, Maeder, and Bogousslavsky (2003) explained the persistence of anomia in two ischemic stroke patients with non-fluent aphasia in the acute stage by the interruption of the dorsolateral pre-frontal circuit projecting from Brodmann areas 9 and 10 to the dorsolateral head of the caudate nucleus.

Striatal and striatocapsular lesions have attracted special attention in the literature given their relationship with the occurrence and persistence of aphasia (Kennedy & Murdoch, 1993; Mega & Alexander, 1994; Weiller et al., 1990). In the present review, 35% of patients with lesions to this region did not present any language disturbances in the acute stage; aphasic symptoms were more marked in patients with lesions extending to the internal capsule, as previously described (Liang et al., 2001). The most common symptoms found were anomia (47% of cases), repetition deficits (35.3%), comprehension impairments (32.3%) and paraphasias (32.3%). Striato-pallidal lesions extending to the internal capsule also resulted in aphasic symptoms in the acute phase (3 cases): anomia (2/3), comprehension deficits (2/3) and paraphasia (1/3) (Carlier et al., 2000; Lieberman, Ellenberg, & Restum, 1986; Puel et al., 1984). Vergara, Verdugo, and Guzmán (1984) reported a patient with a left striatal infarction lesion who presented dysarthria, hypophonia and phonemic paraphasias in the acute stage, while Halkar, Sisterhen, Ammons, Galt, and Alazraki (1997) showed the presence of hypoperfusion in Broca's and Wernicke's areas using SPECT in a patient with striatocapsular lesion and anomic aphasia. However, we found a consistent association only between striatal lesions and dysarthria (Table 5).

We found reports of 26 patients with striatocapsular, striatum (with or without external capsule and claustrum involvement) and striatopallidal (with or without internal capsule involvement) lesions that remained symptomatic after the acute stage. Of these, one patient presented only dysarthria (Milhaud, Magnié, Roger, & Bedoucha, 1994) and two presented only stuttering (Caplan et al., 1990; Carlier et al., 2000). Hypophonia was described in three patients (Démonet et al., 1992; Fabbro, Clarici, & Bava, 1996; Puel et al., 1984). Overall, seventeen (65%) of the 26 patients had comprehension deficit (5 with striatopallidal, 3 with striatal, and

9 with striatocapsular lesions). Eleven patients (38.4%) had repetition deficit. Twenty-three patients (88.4%) had anomia. Paraphasias were described in only 8 patients. Language disturbances were thus found in 23 patients (21 ischemic and 2 hemorrhagic strokes) – in 10 cases, the main hypothesis of the authors was that disturbances were due to hypoperfusion in classic language areas (de Boissezon et al., 2005; Démonet et al., 1992; Radanovic & Scaff, 2003; Wallesch, 1985; Weiller et al., 1990; Willmes & Poeck, 1993). The case reported by Damasio et al. (1982) had a left ICA lesion. Two neurophysiological studies (P600) highlighted deficits in syntactic processing (parsing) affecting auditory comprehension (Friederici et al., 1999; Kotz, Frisch, von Cramon, & Friederici, 2003). One case described by Lieberman et al. (1986) had a lesion extending to the frontal PVWM. The remaining studies discussed previously presented theories (primary involvement of basal ganglia in lexical selection processes, disconnection syndromes, associated cortical hypoperfusion), but none of the studies employed an appropriate methodology for testing the superiority of one hypothesis over the other (De Renzi, Colombo, & Scarpa, 1991; Fabbro et al., 1996; Kennedy & Murdoch, 1993; Robin & Schienberg, 1990). Thus, in at least 50% of cases, evidence suggests that more persistent disturbances were due to cortical hypoperfusion in the classic language areas, whereas in the remaining cases this hypothesis cannot be ruled out due to lack of information.

Isolated lesions of the caudate nucleus had a weak association with non-fluent aphasia and repetition deficits (see Table 5). Of the 36 patients with caudate nucleus lesions extending to the left internal capsule or otherwise, and examined in the acute stage, 22 (61.1%) did not present language disturbances. Five (13.8%) patients were described as having non-fluent aphasia and 8 (22.2%) fluent aphasia; 6 (16.6%) had comprehension impairment, 4 (11.1%) repetition deficits and 14 (38.8%) anomia. Paraphasias occurred in 4 (11.1%) patients and stuttering in one individual. There was one case of isolated dysarthria (Milhaud et al., 1994). Seven symptomatic patients suffered hemorrhagic stroke, where severity of symptoms was proportional to the extent of the stroke, and several patients presented changes in level of consciousness in the first few hours or days post-stroke, suggesting that part of the symptomatology can be explained by mass effect (Cambier, Elghozi, & Strube, 1979; Cappa et al., 1983; Fromm et al., 1985; Krishnan, Tiwari, Pai, & Rao, 2012; Robin & Schienberg, 1990; Stein et al., 1984). Another six patients suffered ischemic infarcts; in two cases the lesion was limited to the territory of the recurrent artery of Heubner (Kumral, Evyapan, & Balkir, 1999; Mega & Alexander, 1994), while the exact site of the lesion could not be determined in one case (Mehler, 1987); in three cases, there was full recovery (Mega & Alexander, 1994; Milhaud et al., 1994; Sonobe, Yashima, Takahashi, Katayose, & Kumashiro, 1991), one remained aphasic (Kumral et al., 1999), and two were not submitted to follow-up assessment (Caplan et al., 1990; Mehler, 1987).

Ten patients with caudate lesions (either isolated or combined with internal capsule) were described as symptomatic when evaluated 1 month after stroke. Three were described as fluent and three as non-fluent; comprehension deficits were found in 2 cases, repetition was affected in 1 case, anomia was found in 4 cases, and paraphasias in 2 patients. One patient had dysarthria only (Pedraza, Donoso, Poblete, & González, 1991). Two persistently aphasic patients showed cortical hypoperfusion in the posterior middle-inferior temporal gyri after ischemic infarcts (Metter et al., 1983), and diffusely (Radanovic et al., 2004). Three patients with hemorrhagic strokes remained symptomatic, in follow-up assessment in up to three months (Cambier et al., 1979; Fromm et al., 1985; Robin & Schienberg, 1990). In these cases, as well as in a case of infarct caused by occlusion of the recurrent artery of Heubner (Kumral et al., 1999), the authors postulated that the

language disturbances could be explained by disconnection between the caudate nucleus and the frontal cortex.

Regarding pallidal lesions, we found 7 descriptions of patients with lesions in the globus pallidus alone or associated with left internal capsule. Of these, 4 had no language disturbances (pure pallidal lesions). Of the three remaining patients (all with associated lesions of the internal capsule and lesions limited to the territory of the anterior choroidal artery), two were described as having transcortical motor aphasia (TMA), and one with fluent aphasia with deficits in naming, semantic processing and verbal fluency. The two TMA cases were reassessed at 8 and 6 months after stroke and reported as having “minimal language problems” and “mild aphasia with dysarthria”, respectively. The author concluded that the symptoms were due to interruption of the cortex-striatum-pallidum-ventral thalamus-frontal cortex loop (Wallesch, 1985). In the remaining cases, no follow-up exam was performed, and the authors interpreted the executive and semantic impairments of the patient as being due to anterior thalamus-frontal cortex disconnection (Mega & Alexander, 1994).

This review identified 17 cases of left internal capsule lesions submitted to language assessment. Of these individuals, 12 tested normal in the acute stage and two had dysarthria only. The remaining 3 patients showed comprehension and naming deficits (one of which was classified as Wernicke's aphasia), 2 had repetition deficit, 1 dysarthria and 2 oral apraxia associated with linguistic alterations. Three patients were reported as chronic aphasics: one presented mild repetition deficit only (associated with impairments in attention and executive functions) (Radanovic et al., 2004), one had only mild naming deficits, whereas one was severely impaired with aphasia resembling Wernicke's aphasia (Robin & Schienberg, 1990). One aphasic patient suffered complete occlusion in the cervical segment of the left ICA, with both anterior and middle cerebral arteries being filled via the right ICA (Lieberman et al., 1986); this might account for some degree of peri-ictal cortical hypoperfusion if the anastomotic pathway was not sufficiently pervious. In the remaining cases, there was no evidence to support cortical involvement, as all lesions appeared to be confined to the anterior choroidal artery territory (Puel et al., 1984; Radanovic et al., 2004; Robin & Schienberg, 1990); in one case, the lesion extended to the adjacent anterior and lateral periventricular white matter (Tanridag & Kirshner, 1985), but this finding was inconclusive, as the patient performed only a CT scan.

Regarding left putaminal lesions (isolated or plus internal and/or external capsule), 43 patients were evaluated in the acute stage of stroke. Of these, 10 (23.2%) were normal, 4 (9.3%) were dysarthric only, and one left-handed subject had dysprosody. Seventeen patients (39.5%) were described as non-fluent, and 10 (23.2%) as fluent. Twelve (27.9%) had comprehension deficits, 13 (30.2%) repetition deficits, and 26 (60.4%) had some degree of anomia. Paraphasias were found in 13 patients (30.2%), dysarthria associated with language deficits was reported in 10 (23.2%). Putaminal lesions were associated with the presence of paraphasias (see Table 6). In seven cases of ischemic infarcts presenting with language problems in the acute stage, concurrent hypoperfusion involving language areas was documented (de Boissezon et al., 2005; Giroud, Lemesle, Madinier, Billiar, & Dumas, 1997; Takahashi et al., 1997). Another five patients with infarcts and acute language disturbances were not submitted to perfusion studies, and their deficits were attributed to a disconnection/deafferentation syndrome involving the fronto-striato-pallido-thalamo-cortical loop (Pulicino, Lichter, & Benedict, 1994; Robin & Schienberg, 1990; Wallesch, 1985). Finally, fifteen patients had hemorrhagic strokes (Alexander & LoVerme, 1980; Alexander et al., 1987; Cappa et al., 1983; Hua et al., 2001; Vergara et al., 1984; Krishnan et al., 2012), and in these cases, compression of

adjacent structures (either by blood clot, edema, or both) is the most probable cause of acute dysfunction.

Twenty-six patients underwent language evaluations after the acute stage: 17 (65.3%) presented comprehension deficits, 13 (50%) repetition deficits, 21 (80.7%) had anomia, paraphasias occurred in 10 patients (38.4%); 14 (53.8%) were dysarthric and 10 (38.4%) had hypophonia. Of these patients, 5 had SPECT or PET studies showing hypoperfusion or hypometabolism involving cortical language areas (Giroud et al., 1997; Metter et al., 1983; Radanovic et al., 2004), one had left fronto-parietal involvement on CT scan (Vergara et al., 1984), and in one case the infarction affected the territory of lateral lenticulostriate arteries (Wallesch, 1985). For most of the remaining patients, persistent deficits were again attributed to a disconnection/deafferentation syndrome involving the fronto-striato-pallido-thalamo-cortical loop (Alexander & LoVerme, 1980; Pedraza et al., 1991; Pullicino et al., 1994; Robin & Schienberg, 1990; Vergara et al., 1984). Exceptions were the cases described by Hua et al., 2001 and Kotz et al. (2003), but these authors focused on clinical aspects related to writing and syntactic disturbances, respectively. Finally, Willmes and Poeck (1993) also held that persistent language symptoms in subcortical lesions were likely due to cortical derangement.

The occurrence and significance of blood flow changes in patients with subcortical lesions have been addressed by a number of different authors. Perani et al. (1987) highlighted the presence of cortical hypoperfusion in patients with subcortical lesions, correlating its occurrence with the presence of language disturbances and neglect, and supported the notion that cortical dysfunction is a crucial factor for the development of cognitive disorders. Démonet, Puel, Celsis, and Cardebat (1991) also described the associated between aphasia and frontal and postero-temporal hypoperfusion in patients with strictly subcortical lesions. In a review on the subject, Metter (1992) described different patterns of cortical hypometabolism in patients with subcortical lesions, based on their series of previous studies (Metter et al., 1983, 1988): hypometabolism in the temporoparietal region in posterior lesions of the putamen, internal capsule and thalamus; frontal hypometabolism in more anterior lesions of the putamen, internal capsule and caudate nucleus; frontal and temporoparietal hypometabolism in more extensive lesions of the lentiform nucleus and internal capsule; and global hemispheric hypometabolism in patients with thalamic lesions. Two further observations in this series of studies included the role of internal capsule lesion in the development of impairments in the control and production of speech, and of the caudate nucleus as an element involved in the motor planning of speech, in association with Broca's area.

At this point, it is important to note that vascular lesions from ischemic and hemorrhagic etiologies produce different local effects, and somewhat distinct pathophysiological mechanisms are to be expected for each etiology. Ischemic infarcts produce direct anoxic neuronal injury in the affected arterial territory, and occlusion of major arteries (such as MCA) may lead to the aforementioned large-scale hypoperfusion that may compromise not only cortical areas but other language-related structures such as white matter pathways, thus originating a disconnection syndrome between language areas. Hemorrhagic strokes also produce direct neuronal damage, but tend to produce an elevated intracranial pressure as a result of mass effect, albeit localized, and this may also lead to ischemic injury in the surrounding structures, including the cortex and its connecting white matter pathways (Nadeau, 2008).

Finally, with regard to the other subcortical lesion models, as neurodegenerative diseases, current evidence in Parkinson's and Huntington's disease indicates that the cerebral cortex is affected diffusely in these clinical conditions, explaining the impairments in language and other cognitive functions found in several studies.

Those language impairments are mostly related to comprehension and syntactic abilities, verbal fluency, and confrontation naming. (Azambuja et al., 2012; Halliday, Leverenz, Schneider, & Adler, 2014; Montoya, Price, Menear, & Lepage, 2006). Language disturbances have also been described in corticobasal degeneration (CBD), including naming deficits, impaired comprehension of grammatically complex sentences, and nonfluent/Broca-like aphasia (McMonagle, Blair, & Kertesz, 2006). Similar findings have been reported in progressive supranuclear palsy (PSP), and clinicopathological studies have revealed that CBD and PSP can present as progressive apraxia of speech and nonfluent aphasia (Josephs & Duffy, 2008).

An evident limitation in the study of language alterations related to basal ganglia lesions derives from the very fact that the function of these structures is not yet fully understood. The basal ganglia participate in learning and selection of cognitive and behavioural adaptive routines, as opposed to non-adaptive ones (procedural skills). The putamen has been most established as involved in motor function. The caudate nucleus receives projections from the supplementary motor area and cingulate gyrus (via subcallosal fasciculus), which suggests a role in action initiation, which includes behaviors and limbic aspects related to speech production (Kemmerer, 2015; Longworth, Keenan, Barker, Marslen-Wilson, & Tyler, 2005). Functional connectivity studies by tractography constitute a promising field to elucidate some challenging issues related to the function of subcortical structures in language.

Also, the hodotopic cognitive model proposed by Duffau (2012) may provide new insights into the understanding of the relationships between subcortical structures and language by proposing a more relevant role for axonal connectivity between language-related areas, and also between language and other cognitive functions. This model explains semantic, phonological and syntactic processing in terms of multiple interconnected large-scale, delocalized sub-networks. Such approach allows the incorporation of subcortical structures in "parallel" pathways in relation to the classic areas of language, and is in agreement with the long-established concept that subcortical lesions involving the periventricular white matter tend to produce more severe and long-lasting aphasic symptoms than those in which it is spared (Duffau, Moritz-Gasser, & Mandonnet, 2014).

6. Conclusions

The current consensus in the literature is that aphasic symptoms after vascular lesions limited to the basal ganglia are fundamentally due to:

- (a) impairments in cortical circulatory dynamics, i.e., secondary to ischemia/hypoperfusion in cortical territories dependent on the MCA and that have poor collateral circulation, or to compression with consequent infarction of adjacent cortical structures, in cases of larger hemorrhagic lesions (Hillis et al., 2002, 2004; Nadeau & Crosson, 1997; Weiller et al., 1993).
- (b) disconnection syndrome: proximal occlusions in the MCA can cause ischemia of the white matter pathways connecting the thalamus to the fronto-subcortical-temporal language network, causing additional aphasic symptoms secondary to thalamic disconnection (Nadeau, 2008). In addition, a number of authors maintain that only striatocapsular lesions can produce a clinical picture sufficiently consistent and persistent to be defined as "subcortical aphasia" (Alexander & Hillis, 2008; Ardila, 2014).

Limitations of this review include the fact it was not possible to determine the incidence or prevalence of language disturbances

due to basal ganglia lesions, as most studies enrolled in the review were multiple case reports; most studies addressing the frequency of occurrence of cognitive deficits after subcortical lesions did not provide a description of symptoms for each patient, but rather pointed to the proportion of occurrence of a given deficit (for instance, “23% of patients had aphasia”). Moreover, papers were highly heterogeneous regarding study design, inclusion and exclusion criteria, language and neuropsychological assessment, and aim, thus rendering collection of data laborious and often incomplete, which, in turn, hampered more refined statistical analyses. Finally, a great proportion of studies enrolled patients with cortical and/or PVWM involvement, and had to be excluded. Nevertheless, we believe that this review may contribute to provide an overall view of the relationship between pure basal ganglia lesions and language symptoms. As for future perspectives on the issue of subcortical aphasia, further studies using diffusion tensor tractography may contribute to elucidate the role of basal ganglia structures in language symptoms.

Conflict of interest

The authors report no conflict of interest.

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Appendix A. Supplementary material

Supplementary data associated with this article can be found, in the online version, at <http://dx.doi.org/10.1016/j.bandl.2017.05.003>.

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